

Reviewer-first figure architecture for the DM1 thyroid-lineage state manuscript. This is a rebuild plan, not author-voice protected prose.

ARTICLE ARCHITECTURE

A reviewer-first rebuild of the DM1 thyroid-lineage state manuscript

The current manuscript should be reconstructed around five figures, where each figure answers one predictable reviewer attack: artifact, biology, mechanism, validation and clinical relevance.

Prepared 8 July 2026. Scope: figure architecture, reviewer defense and submission package triage. Voice-protected manuscript sections remain author-only.

Editorial verdict

The paper should stop behaving like a repository of every successful analysis. Nature Communications needs one clean claim ladder, with side analyses moved into Extended Data or Supplementary Information.

Weak framing

"Here is an 8-gene panel that predicts RAI failure."

This invites attacks on panel selection, retrospective endpoints, clinical overclaim and lack of wet-lab causality.

Submission-grade framing

"A hidden thyroid-lineage state exists in papillary thyroid cancer, is not explained by canonical drivers, converges with known dedifferentiation biology, has an epigenetic correlate, reproduces across cohorts and is positioned for prospective RAI harm-avoidance validation."

Reviewer attack map

Predictable attack	Main figure that must answer it	Required answer
"Your eight genes created the cluster."	Figure 1	Pan-genome recovery and driver-only failure must appear before biology.
"This is just BRAF/RAS/fusion biology."	Figures 1 and 3	Driver AUC near chance, cross-driver DM1 prevalence and methylation structure.
"What does DM1 mean biologically?"	Figure 2	Thyroid-lineage and iodine-handling loss, not just clustering.
"Methylation is only correlative."	Figure 3	State explicitly: epigenetic correlate, not causal proof.
"TCGA-only cherry-pick."	Figure 4	Small external panel: Korean PTC, advanced thyroid, proteomics and compressed forest.
"Clinical claim is too strong."	Figure 5	Retrospective outcome and deployability, with prospective-validation boundary.

Main claim ladder

1. Existence DM1/DM2 is a real transcriptomic state, not an eight-gene artifact.
2. Biology DM1 means coordinated loss of thyroid lineage and iodine-handling machinery.
3. Mechanism correlate DM1 shows promoter hypermethylation of lineage genes across driver contexts.
4. Validation The state recurs across independent cohorts and modalities.
5. Clinical relevance DM1 aligns with adverse outcome, RAI-failure biology and FFPE deployability.

New five-figure architecture

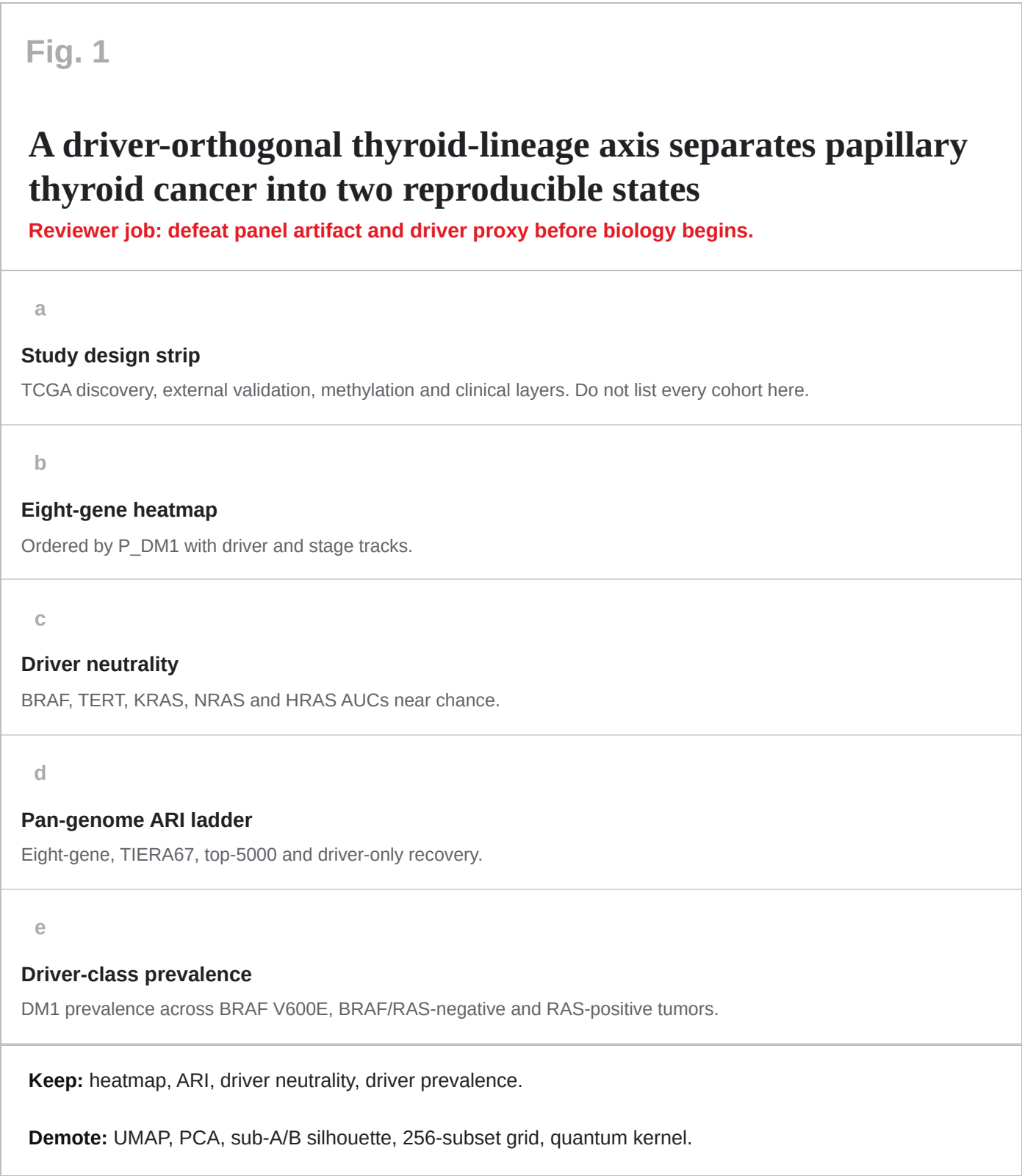


Fig. 2

DM1 represents coordinated loss of thyroid lineage and radioiodine machinery

Reviewer job: explain what the state means biologically.

a

Panel-gene expression

TF3 plus effector5 expression shift between DM1 and DM2.

b

Pathway enrichment

Thyroid transcription factor and iodine-handling pathway depletion.

c

TDS-16 convergence

Panel-8 as compact readout of canonical differentiation loss.

d

Landa overlap

PDTC/ATC lineage silencing convergence, including 5 of 8 genes.

e

Proteomic direction

Mun 2025 direction consistency, used only if clean enough for main.

Keep: compact biology and known dedifferentiation convergence.

Demote: detailed single-cell, deconvolution, pseudotime and pathway walls.

Fig. 3

DM1 is associated with promoter hypermethylation of thyroid-lineage genes across driver contexts

Reviewer job: give mechanistic depth without claiming causality.

a

HM450 heatmap

Promoter beta values for eight genes plus controls.

b

Mean beta shift

Eight-gene promoter methylation difference between DM1 and DM2.

c

Beta-expression bridge

TPO, DIO1, TSHR and TG scatter linking methylation to expression.

d

Driver-context beta

Per-driver methylation distribution to avoid a single-driver mechanism story.

e

Boundary inset

"Epigenetic correlate, not causal proof" placed inside the figure.

Keep: methylation only, with cautious language.

Demote: PRISM, DepMap, HMA trial schematic, spatial and pan-cancer analyses.

Fig. 4

The DM1 axis recurs across independent PTC, advanced-thyroid and proteomic cohorts

Reviewer job: defeat TCGA cherry-pick with a small interpretable validation panel.

a

Lee 2024 Korean PTC

Large independent PTC validation, shown as one clean distribution panel.

b

Landa advanced thyroid

Advanced-end alignment with PDTc/ATC dedifferentiation.

c

Mun proteomics

Proteomic direction consistency across measurable panel proteins.

d

Four-entry forest

TCGA, Lee, Landa and Mun only. The full forest goes to Extended Data.

e

Supplement pointer

Full 14-cohort validation and 80/80 matrix summarized without crowding main text.

Keep: one cohort per validation class.

Demote: K2 calibration-heavy panels, GPL570 details, K-Thyro and full validation atlas.

Fig. 5

DM1 is associated with adverse outcome, RAI-failure biology and FFPE deployability

Reviewer job: show why this matters clinically without treatment-selection overclaim.

a

OS forest

Pooled TCGA and MSK survival association.

b

Multivariable Cox

Adjusted model including available age, stage and TERT covariates.

c

Post-RAI refractory alignment

Retrospective RAI-failure biology with n and caveat visible.

d

ATA uncertainty zone

Intermediate-risk enrichment or decision-uncertainty mosaic.

e

FFPE portability

Fresh frozen versus FFPE concordance for deployment plausibility.

Keep: retrospective relevance and deployability.

Demote: selpercatinib waterfall, prospective trial schema, decision curve, time-dependent ROC and calibration.

Extended Data package

Extended Data	Content	Reason
ED1	Full ordered heatmap with all annotation tracks	Detail for Figure 1.
ED2	Panel-size, leave-one-out and ARI sensitivity	Defense against "why eight genes."
ED3	Single-cell thyrocyte-intrinsic validation	Defense against stromal or immune confounding.
ED4	Within-DM1 fusion-positive versus fusion-negative methylation	Fusion-independent methylation defense, but small n.
ED5	Full cross-cohort validation atlas plus 80/80 matrix	Defense against cherry-pick.
ED6	Survival diagnostics: KM, ROC, calibration and PH checks	Clinical-statistics support.

What must be cut from the main story

Main-text standard

- One figure equals one claim.
- Every panel answers a reviewer attack.
- Clinical language remains retrospective and prospective-validation bounded.
- Mechanism language says "epigenetic correlate", not causality.

Demote or remove

- Spatial, PRISM, DepMap and pan-cancer analyses.
- Sub-A/B, quantum kernel and algorithm-flex panels.
- Full 14-cohort forests inside the main figure.
- Therapy schematics that imply immediate treatment selection.

Reviewer-grade central claim

We identify a driver-orthogonal thyroid-lineage state in papillary thyroid cancer that is recovered by pan-genome expression structure, corresponds to coordinated iodine-handling loss, has an epigenetic silencing correlate, recurs across independent cohorts and modalities, and is associated with retrospective RAI-failure biology and adverse outcome.

Do not claim: "DM1 predicts RAI response and should guide treatment." That is prospective-validation territory.
